

JEV-1.13 on High School U.S. History: Test Report

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Abstract

This report presents the results of the model JEV-1.13 on the high school U.S. history subject of the MMLU benchmark. The test comprises 188 four-option multiple-choice questions; the model answered 180 correctly, an accuracy of 95.7%, far above the 25% random-guessing level. The model also states a probability for every option: the probability assigned to the chosen option (the chosen probability) was 1.00 for 134 questions, which were almost all correct (99.3%), and below 1.00 for 54 questions, which were 87.0% correct. The whole test consumed 122 404 input tokens and 8 460 output tokens, costing USD 0.0051 in total. In this scenario the model answers accurately at very low cost, and its chosen probability indicates how reliable an answer is.

1 Dataset

The data come from the test split of the high school U.S. history subject of the MMLU (Massive Multi-task Language Understanding) benchmark [1]. The test consists of 188 four-option single-answer questions. Each item presents a passage of material (a quoted document or a situational description) followed by a question; the model must select the one correct option. The answer keys provided by the dataset are distributed nearly uniformly across the four letters (A 47, B 42, C 51, D 48). The subject under test is the model JEV-1.13 (identifier typesafe/jev-1.13-20260917).

2 Results

All questions are scored against the answer key provided by the dataset.

2.1 Overall Results

Table 1 summarizes the overall outcome. Of the 188 questions, the model answered 180 correctly, an accuracy of 95.7% (95% confidence interval 91.8%–97.8%). Random guessing on four-option questions yields 25% on average, so the model scores far above chance.

Table 1: Overall results

Item	Value
Questions	188
Correct	180
Accuracy	95.7%
95% confidence interval	91.8%–97.8%

2.2 Chosen Probability and Accuracy

Besides the selected option, the model states a probability for each of the four options; the probability assigned to that option is called the chosen probability.

For 134 questions (71.3%) the chosen probability was 1.00, and 99.3% of them were answered correctly; for the remaining 54 questions it was below 1.00, with 87.0% accuracy. The chosen probability averages 0.99 on the 180 correct answers and 0.83 on the 8 wrong ones.

Figure 1 groups questions by chosen probability. Only 4 of the 7 questions below 0.90 were correct; accuracy rises across the bands overall and exceeds 99% in the 0.99 and 1.00 bands. The chosen probability therefore separates reliable answers from unreliable ones and can serve as a reference for deciding whether to trust an individual answer.

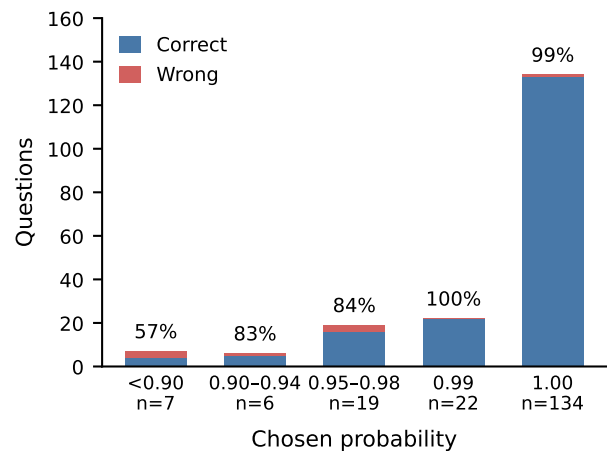


Figure 1: Question counts and accuracy by chosen-probability band. Bar height is the number of questions in the band (blue: correct, red: wrong); the percentage above each bar is the accuracy of that band.

2.3 Cost

As Table 2 shows, the whole test consumed 122 404 input tokens and 8 460 output tokens, costing USD 0.0051 in total.

Table 2: Cost of the test

Item	Amount
Input tokens	122 404
Output tokens	8 460
Total cost	USD 0.0051

3 Conclusion

On 188 high school U.S. history multiple-choice questions, JEV-1.13 answered 180 correctly, an accuracy of

95.7%, far above the random level. More than seventy percent of its answers had a chosen probability of 1.00 and were almost all correct, while answers below 0.90 were correct less than sixty percent of the time. The entire test cost about half a U.S. cent. Overall, in this scenario JEV-1.13 answers accurately at very low cost, and its chosen probability can be used to flag unreliable answers.

References

- [1] Dan Hendrycks, Collin Burns, Steven Basart, Andrew Critch, Jerry Li, Dawn Song, Jacob Steinhardt. *Measuring Massive Multitask Understanding*. ICLR 2021. Data taken from the cais/mmlu test split.